Data Analytics for Business

Homework 5 – Part 2

Collaborative Filtering

In this homework, we will work on an exercise to help you better understand the concepts involved in collaborative filtering. We will then use a R package to perform collaborative filtering based on a real data set of movie ratings and recommendations.

The data set used for this homework, **MovieLense**, is included in the **recommenderlab** package.

Task 1: Calculations

Let's first consider a simple example of user rating data as follows.

	I_1	I_2	I_3
U_1	2		3
U_2	5	2	
U_3	4	4	1

1. Applying the algorithm of **User-based Collaborative Filtering (UBCF)** as we learned in class, manually calculate the predicted rating of User 1 (U_1) on Item 2 (I_2).

   Hints: here are the steps you should follow:

   i) Normalize the table by subtracting user-mean;

   ii) Calculate the similarities between U_1 and U_2 (s_{12}) and between U_1 and U_3 (s_{13});

   iii) Calculate the predicted rating as the weighted average of U_2's and U_3's ratings on I_2, using s_{12} and s_{13} as the weights.

   Write down your entire calculation process, especially the formulas for Steps ii) and iii), along with the calculation results.

Task 2: Apply to MovieLense Data

Next let's perform collaborative filtering on a real data set. We will use the **MovieLense** data, which is included in the **recommenderlab** package. The data was collected through the MovieLens web site (movielens.umn.edu) during the seven-month period from September 19th, 1997 through April 22nd, 1998. The data set contains about 100,000 ratings (1-5) from 943 users on 1664 movies. Movie and user metadata is also provided in *MovieLenseMeta* and *MovieLenseUser*.

2. Install the **recommenderlab** package and load it into R. (If prompted, you may need to install a few packages used in the **recommenderlab** package.) Load the **MovieLense** data (using **data()** function). To perform personalized recommendations, based on your own preferences, rate the following 15 movies (all of which are included in the **MovieLense** dataset). First create your ratings as a matrix, and then convert it into a "**realRatingMatrix**".

Row Index	Movie Name	Row Index	Movie Name
1	Toy Story (1995)	64	Shawshank Redemption, The (1994)
11	Seven (Se7en) (1995)	71	Lion King, The (1994)
12	Usual Suspects, The (1995)	82	Jurassic Park (1993)
22	Braveheart (1995)	88	Sleepless in Seattle (1993)
23	Taxi Driver (1976)	94	Home Alone (1990)
29	Batman Forever (1995)	96	Terminator 2: Judgment Day (1991)
50	Star Wars (1977)	98	Silence of the Lambs, The (1991)
56	Pulp Fiction (1994)		

3. Perform **UBCF** on the **MovieLense** data. Then use your own ratings as the new data, and predict a "**topNList**" for you. You can set ***n*=10** (which is the default value), and display the list. This is a list of the top 10 most recommended movies specially for you!